Team Name: Commit and pray

Gradient-Guided Selective Untying in Small Language Models

Name	Section	SRN
Greeshma G	H	PES1UG24AM462
Sindhu K S	H	PES1UG24AM433
Sanjith Suresh Kumar	E	PES1UG24AM250
Shriyadita Mohanty	E	PES1UG24AM270

Problem Statement and Proposed Extension

Weight tying shares token vectors between input encoding and output prediction, reducing parameters [1]. Recent work reports that output-side gradients can dominate tied embeddings during early training [3], but its intervention does not establish consistent downstream gains. We therefore test whether a targeted, token-level intervention can help: rather than fully untying the vocabulary, we selectively add output-side parameters to tokens with strong input/output gradient conflict. The SLM is pretrained from scratch and instruction-tuned.

Model architecture and training. The decoder-only Transformer uses 6 pre-LayerNorm blocks, hidden size 384, 6 causal attention heads, GELU feed-forward width 1,536, learned positional embeddings, an 8,000-token BPE vocabulary and context length 256 (approximately 13.8M parameters). Pretraining uses next-token cross-entropy; supervised fine-tuning (SFT) uses teacher forcing and response-only cross-entropy. All parameters remain trainable with AdamW and mixed precision.

Proposed selective untying procedure

Before SFT, use a calibration phase on SFT training data to estimate input- and output-side gradient contributions for each eligible vocabulary row. We propose mean negative cosine similarity as a directional-conflict score; aligned pairs score 0. This adapts the negative-cosine conflict signal used in gradient-conflict methods [5,6] to the two paths of a tied embedding. Exclude low-occurrence rows and near-zero gradients because their estimates may be unreliable. Select up to 400 rows, capped at 10% of eligible rows. For selected rows, use $E_{out,i} = E_i + \text{Delta}_i$, with Delta_i initialized to zero, analogous to the zero-update initialization principle used in LoRA [7], so the initial logits match the tied model. The shared E_i remains trainable and still receives both gradient contributions. This token-level selection plus output-offset combination is our proposed heuristic, not an established method in the cited papers. Pilot experiments fix thresholds and check ranking stability.

Datasets and Sources

Pretraining - TinyStories. Use TinyStories-train.txt and TinyStories-valid.txt from roneneldan/TinyStories. Fit BPE on training text only; pack sequences with end-of-story delimiters.

Instruction tuning - TinyStoriesInstruct. Use TinyStories-Instruct-train.txt and TinyStories-Instruct-valid.txt from roneneldan/TinyStoriesInstruct. Parse complete end-of-text-delimited records. Convert the Words field into a keyword-constrained story instruction; use only the Story body as the response. Retain references containing all requested words.

Splits and preprocessing. Hold out development stories from instruction training; use official instruction validation for testing. Remove evaluation-story overlaps from both training corpora before pretraining. Retain complete SFT pairs within 256 BPE tokens; report filtering counts. Calibration uses training data only. Freeze preprocessing after the pilot.

Evaluation Criteria

Compare tied embeddings, conflict-selected offsets, frequency-matched random selection, frequency selection, gradient-norm selection and full untying. Match selective parameter budgets, SFT tokens, hyperparameters and decoding; use one pretrained checkpoint, fresh optimizers and three paired SFT seeds. Measure all-keyword satisfaction by case-insensitive whole-word matching, response negative log-likelihood, pretraining perplexity, repetition and blinded coherence/keyword-integration ratings.

Pre-specified success criterion: count the method as a positive result only if mean all-keyword satisfaction is at least 2 percentage points above both frequency and frequency-matched-random baselines across the three paired seeds, with no mean degradation in coherence rating or response NLL relative to the tied baseline. Otherwise report the result as mixed/null rather than treating any directional improvement as success.

To test whether conflict selection adds information beyond frequency, all selective baselines use the same eligible pool and parameter budget. Report selection overlap/Jaccard, conflict-frequency rank correlation, and performance within frequency bins. Report paired differences, uncertainty and GPU time. A throughput pilot will measure pretraining, calibration and representative SFT time and project the full 18-run GPU cost; the approximately 50 A10-hour figure is a budget cap, not an assumed runtime. Blinded rating time is tracked separately. Findings are conditional on this checkpoint and constrained story generation.

Expected Contribution and Learning Goals

The contribution is a proposed token-level conflict-guided parameter-allocation heuristic for selective untying. The study tests whether it outperforms simple selectors under matched budgets; no improvement is assumed beforehand. Learning goals include Transformer training, gradient decomposition and controlled evaluation.

Reference Papers and Sources

[1] Press, O. & Wolf, L. (2017). *Using the Output Embedding to Improve Language Models.* aclanthology.org/E17-2025/
[2] Eldan, R. & Li, Y. (2023). *TinyStories: How Small Can Language Models Be and Still Speak Coherent English?* arxiv.org/abs/2305.07759
[3] Lopardo, A., Harish, A., Arnett, C. & Gupta, A. (2026). *Weight Tying Biases Token Embeddings Towards the Output Space.* arxiv.org/abs/2603.26663
[4] Chung, H. W., Févry, T., Tsai, H., Johnson, M. & Ruder, S. (2020). *Rethinking Embedding Coupling in Pre-trained Language Models.* arxiv.org/abs/2010.12821
[5] Shi, G., Li, Q., Zhang, W., Chen, J. & Wu, X.-M. (2023). *Recon: Reducing Conflicting Gradients from the Root for Multi-Task Learning.* arxiv.org/abs/2302.11289
[6] Yu, T., Kumar, S., Gupta, A., Levine, S., Hausman, K. & Finn, C. (2020). *Gradient Surgery for Multi-Task Learning.* arxiv.org/abs/2001.06782
[7] Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L. & Chen, W. (2021). *LoRA: Low-Rank Adaptation of Large Language Models.* arxiv.org/abs/2106.09685